Report on Script

Digital Sovereignty for Kashmir

July 12, 2026

1 Section 0 — Why This, Why Now

Slide 1 — Title [0:25 | running total 0:25]

On screen: Title, subtitle, your name, "Open Library Kashmir (OLK)", July 2026.

Let Me Explain

Good evening, everyone. Thank you for coming. Tonight I want to talk about something we've touched on before in smaller conversations — building our own digital infrastructure, right here, so that our data, our platforms, and our future don't have to live on someone else's servers, under someone else's laws. I'll walk through why this matters, what it actually takes technically, and end with a real, fundable first step we could start this year.

Slide 2 — Tonight's Roadmap [0:45 | running total 1:10]

On screen: The table of contents: the six sections plus "The Hope".

Let Me Explain

Here's the shape of tonight's talk. We'll start with the "why" — the motivation. Then we'll get practical: the internet and electricity we'd need, the physical room the servers live in, starting with a VPS as our on-ramp, the hardware for scaling to a hundred thousand users, and the software and skills we'd need to run it. And I'll close with a proposal — a modest, real pilot we could begin, not a someday dream.

Slide 3 — Divider: Why This, Why Now [0:10 | running total 1:20]

On screen: Section title only.

Let Me Explain

So — why this, why now.

Slide 4 — The cost of renting our own knowledge [1:35 | running total 2:55]

On screen: Bar chart of global cloud market share (AWS 30%, Azure 24%, Google 11%, others 35%); gold takeaway box on "whose infrastructure, whose law".

Terms to know:

CLOUD Act A 2018 US law: any company headquartered in the US must hand government-requested data to US authorities, even if the server holding it is physically in another country.

Hyperscaler Industry term for a cloud provider operating at massive, global scale (AWS, Azure, Google Cloud, Alibaba Cloud, etc.).

Let Me Explain

Let's start with an uncomfortable fact. Roughly two-thirds of the world's cloud infrastructure spend sits with three American companies — Amazon, Microsoft, and Google. And every one of them is bound by something called the CLOUD Act — a US law that says a US company must hand over data on request, no matter which country the actual server sits in. So when we say our data is "hosted in India" on an American platform, that's a comforting label — it is not sovereignty. The question was never whether we depend on cloud infrastructure. We already do, every day. The real question is: whose infrastructure, under whose law, and who profits from it.

Slide 5 — OLK today: a working example, and its ceiling [1:20 | running total 4:15]

On screen: Two columns: "What we already built" vs. "The ceiling".

Terms to know:

Next.js / React The open-source web framework OLK's interface is built with.

PostgreSQL The open-source database engine that stores every book, user, and request.

Supabase An open-source platform that bundles Postgres with authentication, file storage, and an API layer — what OLK currently runs on.

`ap-south-1` Amazon's internal code for its Mumbai data center region.

Let Me Explain

Now, here's where we already stand. Open Library Kashmir — OLK — is real. People are donating books, lending them, requesting them, right now. It's built entirely on open-source technology: Next.js and React for the interface, PostgreSQL as the database, all wired together through a platform called Supabase. Every layer of that software, we control. But here's the ceiling: today it physically runs inside Amazon's data center in Mumbai. Being geographically close isn't the same as being legally ours — it's still Amazon's rack, Amazon's power, Amazon's law. The good news is that because everything is open-source, we don't need to rewrite a single line to change that. We just need to redeploy it onto infrastructure we actually own.

Slide 6 — Beyond one app: why this could matter for millions [1:05 | running total 5:20]

On screen: Hub-and-spoke diagram: Open Source Infra at the centre, spokes to libraries, cooperatives, health records, schools.

Let Me Explain

And OLK is not the destination — it's the first baby step. The exact same pattern — a self-hosted database, authentication, storage, a web front end — is the blueprint for every community institution that doesn't yet have a digital backbone of its own: farmer and artisan cooperatives, community health records, local school archives. If we get this right once, we've built a template the whole region can reuse.

2 Section I — Internet & Electricity: The Foundation

Slide 7 — Divider: Internet & Electricity [0:10 | running total 5:30]

Let Me Explain

Let's talk foundations — the internet and the electricity this would actually run on.

Slide 8 — How much internet do we actually need? [1:35 | running total 7:05]

On screen: Bandwidth math worked out step by step; takeaway on CDN/edge caching.

Terms to know:

Gbps / Mbps Gigabits and megabits per second — units of network speed. 1 Gbps = 1000 Mbps.

Concurrent users People actively using the app *at the same instant*, as opposed to total registered users.

CDN / edge cache A "Content Delivery Network" — copies of your content stored on servers closer to users, so they don't have to reach all the way back to the origin server for every request.

Let Me Explain

Let's do some real math. Say we reach a hundred thousand registered users. At any given moment, maybe eight percent are actively online — that's eight thousand concurrent users. If each one is pulling around thirty kilobytes a second — normal for a text-and-image app like ours — that's roughly two hundred forty megabytes a second, or about two gigabits per second, sustained. Add burst headroom for peak load, and we're looking at eight to ten gigabits per second. A typical home or business fiber line here in Srinagar — a hundred megabits to a gigabit — is an order of magnitude short of that. We'd need datacenter-grade, redundant links — multiple carriers, multiple physical paths. And given Kashmir's history with internet shutdowns, a single line isn't just slow, it's a single point of total failure. Our design principle: keep a cache outside the valley — in Hong Kong or Singapore — so people can still read even if a local shutdown stops new writes.

Slide 9 — Electricity: the honest picture [1:20 | running total 8:25]

On screen: Two columns: where J&K's grid stands vs. what that means for our design.

Terms to know:

RDSS Revamped Distribution Sector Scheme — the central government's ongoing power-sector reform program that J&K is implementing.

KPDCL Kashmir Power Distribution Corporation Limited — the utility running the local grid.

Online double-conversion UPS A battery backup that continuously converts incoming power to DC and back to AC, so it can take over with *zero* switchover delay when grid power drops — unlike cheaper "standby" UPS units.

Auto-transfer switch Hardware that automatically switches a building's power source from grid to generator (and back) without a human flipping anything.

Let Me Explain

On power: the grid here is genuinely improving. Billing efficiency has gone from fifty-six percent to seventy-seven percent under the RDSS reform program. But KPDCL still runs scheduled maintenance outages regularly, and unscheduled ones remain common, especially in winter. So our design has to treat grid power as supplementary, not primary, from day one. That means a rack-level online double-conversion UPS — a battery system that instantly and seamlessly takes over the moment grid power drops — sized for at least thirty to sixty minutes of runtime. Behind that, a diesel generator with an automatic transfer switch and enough fuel for a couple of days. Solar and battery can supplement over time, but never as the sole source — Kashmir's winter sun just isn't reliable enough for that.

3 Section II — Physical Environment

Slide 10 — Divider: Physical Environment [0:10 | running total 8:35]

Let Me Explain

Now — where do these machines actually live?

Slide 11 — The room the servers live in [1:20 | running total 9:55]

On screen: Cooling / safety on the left, physical security and a takeaway on the right.

Terms to know:

Free-air / economizer cooling Using outside air directly to cool a server room instead of running mechanical air-conditioning — works whenever the outside air is already cool and dry enough.

FM-200 / Novec Clean-agent fire suppression gases that put out fires without leaving behind water or residue that would destroy electronics.

Positive-pressure filtration Keeping air pressure inside the room slightly higher than outside, so filtered air leaks *out* through any gap rather than unfiltered dust leaking *in*.

Let Me Explain

Kashmir's climate is actually an asset here. Cool air most of the year means we can use free-air cooling — essentially just letting outside air do the work — and only need real air-conditioning during summer peaks and for humidity control. We'd seal the rack and use positive-pressure filtration to keep valley dust and pollen out. For safety, clean-agent fire suppression — never water sprinklers over live electronics — plus smoke and heat detectors wired to automatic shutdown. And physically: one access-controlled room, logged entry, CCTV, locking cabinets. For our pilot, this is genuinely one well-prepared room with a single rack — not a purpose-built data center. That's a later, bigger project.

4 Section III — Starting with VPS: Our On-Ramp

Slide 12 — Divider: Starting with VPS [0:10 | running total 10:05]

Let Me Explain

So how do we actually start? With a VPS — our on-ramp.

Slide 13 — Why VPS first: the staged path [1:05 | running total 11:10]

On screen: Three-stage diagram: VPS → Colocation → Owned facility.

Terms to know:

VPS Virtual Private Server — a slice of a shared physical machine, rented by the hour or month, that behaves like your own private computer.

Colocation ("colo") Renting rack space, power, and network connectivity inside someone else's data center, while the physical server hardware inside it is yours.

Capex Capital expenditure — upfront money spent buying physical assets (as opposed to ongoing rental/subscription costs).

Let Me Explain

Think of this as three stages. Stage one: a VPS — a virtual private server — rented from a Chinese cloud provider. Near-zero upfront cost, and it's where we learn real operations: Linux, backups, monitoring. Stage two: colocation — we rent rack space, power, and network in someone else's facility, but the hardware inside it is ours. That's exactly where our pilot lands. Stage three: our own facility, full physical control, based here in Kashmir, serving the region. Every one of these stages runs the exact same open-source software. Moving between them is a migration, not a rewrite. We're not choosing between a VPS and our own data center — we're choosing the order in which we build confidence and capability.

Slide 14 — The Chinese cloud landscape [1:05 | running total 12:15]

On screen: Four bullets on why Chinese providers specifically.

Let Me Explain

Why Chinese cloud providers specifically? Alibaba Cloud, Tencent Cloud, and Huawei Cloud are three of the largest cloud operators in the world by capacity — mature, globally reachable, and entirely outside the reach of that US CLOUD Act we talked about earlier. All three run international sites in Hong Kong and Singapore built specifically for customers outside mainland China. And Huawei Cloud in particular runs on the same processor and operating system family — Kunpeng and openEuler — that we'd eventually deploy on our own hardware. One skill set covers both stages. And to be direct about it: everything from this point forward in the talk is Chinese-made. No AWS, no Azure, no Google.

Slide 15 — Alibaba vs. Tencent vs. Huawei Cloud [1:20 | running total 13:35]

On screen: Comparison table across four rows; gold recommendation box.

Terms to know:

ECS Elastic Compute Service — Alibaba's (and Huawei's) name for a rentable virtual server.

CVM Cloud Virtual Machine — Tencent's name for the same idea.

Lighthouse Tencent's simplified, all-in-one VPS product aimed at small projects — easier to set up than a full CVM.

Latency The delay for a packet of data to travel from Srinagar to the server and back, measured in milliseconds (ms).

Let Me Explain

Comparing the three: Alibaba Cloud has the broadest footprint outside China and the most mature international billing. Tencent Cloud has a product called Lighthouse — basically a simplified, all-in-one VPS — perfect for a first pilot. Huawei Cloud's advantage is architectural continuity: Kunpeng and openEuler, the same stack as our future hardware. All three have a nearby region in Hong Kong or Singapore, with latency to Srinagar around forty to eighty milliseconds — perfectly usable. My recommendation: start on Tencent Lighthouse or Alibaba's ECS for the cheapest, fastest first pilot, and plan our eventual on-premises hardware around Huawei's ecosystem for continuity.

Slide 16 — Getting a Chinese VPS as a foreign user [1:05 | running total 14:40]

On screen: Five numbered practical steps; gold takeaway box.

Terms to know:

ICP license A mainland Chinese government filing required for any website physically hosted on mainland Chinese servers — avoided entirely by choosing a Hong Kong or Singapore region instead.

Pay-as-you-go vs. reserved instance Paying by actual usage/hour, versus committing upfront to a fixed term (e.g. one year) for a discount.

Let Me Explain

Practically, here's how we'd actually get one of these as outsiders. Register on the international site — not the mainland Chinese site. Choose a Hong Kong or Singapore region — that alone avoids China's mainland ICP licensing requirement, which only applies to websites physically hosted inside mainland China. Identity verification is just a passport and organization documents — we'd register OLK formally to do this cleanly. Payment works with an ordinary international credit card. And critically: start pay-as-you-go on a small instance before committing to any annual plan, so we validate our actual workload first. None of this requires setting up a China-based business. It requires an afternoon of paperwork.

5 Section IV — Hardware for Full Scale: 100,000 Users

Slide 17 — Divider: Hardware for Full Scale [0:10 | running total 14:50]

Let Me Explain

Now let's talk about the hardware we'd need to actually scale to a hundred thousand users.

Slide 18 — Storage 101: what we need, how much [1:20 | running total 16:10]

On screen: SSD vs. NVMe on the left, sizing math for 100k users on the right.

Terms to know:

SATA SSD A solid-state drive connected over the older SATA interface, capped around 550 MB/s.

NVMe A storage protocol that lets SSDs talk directly over the same high-speed PCIe lanes as the processor — much faster than SATA.

IOPS Input/Output Operations Per Second — how many individual read or write operations a drive can handle each second; more important than raw speed for databases doing lots of small operations.

WAL Write-Ahead Log — PostgreSQL's record of every change, written before it's applied, used to guarantee no data is lost in a crash.

Let Me Explain

Two storage technologies matter here. SATA SSDs are fast and cheap but capped around five hundred fifty megabytes a second by their interface. NVMe drives talk directly over the same high-speed lanes as the processor — multiple gigabytes per second, and far more importantly, far higher IOPS, which just means how many small read-and-write operations it can handle per second. That matters enormously for a database like ours doing thousands of tiny random reads and writes. For sizing: at a hundred thousand users, with about five megabytes of data each, plus book cover images, plus database overhead like indexes and write-ahead logs and backups, we land on a design target of two to five terabytes of fast NVMe storage, with a separate cheaper tier for older, archived data.

Slide 19 — RAID: redundancy is not optional [1:20 | running total 17:30]

On screen: Table of RAID 0/1/5/6/10 with fault tolerance and usable capacity; recommendation box.

Terms to know:

RAID Redundant Array of Independent Disks — combining multiple physical drives so data survives one (or more) drive failures, and/or goes faster.

Striping Splitting data across multiple drives so reads/writes happen in parallel — faster, but no protection on its own.

Mirroring Writing the same data to two drives simultaneously — if one dies, the other has an exact copy.

Parity Extra, mathematically-derived data stored alongside the real data, which lets the array reconstruct a lost drive's contents.

Let Me Explain

RAID is how we protect against a hard truth: drives fail. RAID zero just spreads data across disks for speed, with zero protection — if one drive dies, everything's gone. RAID one mirrors two drives — simple, but you lose half your capacity. RAID five and six add parity data so the array survives one or two drive failures respectively. RAID ten mirrors pairs of drives and then stripes across those pairs — it gives you the best of both speed and redundancy, at the cost of losing half your raw capacity. Our pick: RAID ten for the live database — it's the system of record, worth the extra cost. RAID six is fine for the bulk image and backup archive, where raw capacity matters more than speed.

Slide 20 — ECC RAM: why a bit-flip matters [1:05 | running total 18:35]

On screen: Why ECC matters, then sizing guidance for 10–20k and 100k scale.

Terms to know:

ECC RAM Error-Correcting Code memory — detects and automatically fixes single-bit errors as they happen, instead of silently corrupting data.

DIMM Dual In-line Memory Module — the physical memory stick that plugs into a server.

`shared_buffers` A PostgreSQL setting controlling how much RAM the database reserves for its own working cache.

Let Me Explain

Here's something people don't think about: ordinary consumer memory has zero protection against random bit-flips — caused by things like cosmic rays or electrical noise. Rare, yes, but real, and it gets worse the more memory chips you run and the longer they run. ECC — error-correcting code — RAM detects and silently fixes single-bit errors in real time. For a system of record — who owns which book, who borrowed what — a silently corrupted database row is far worse than a slow page load. This is not where we cut costs. For sizing, a hundred twenty-eight gigabytes gives PostgreSQL comfortable room for our ten-to-twenty-thousand-user pilot. Full hundred-thousand-user scale would mean two hundred fifty-six to five hundred twelve gigabytes, or splitting the load across multiple machines instead.

Slide 21 — The Chinese hardware stack, end to end [1:20 | running total 19:55]

On screen: Table: CPU, chassis, storage, OS, networking, all Chinese vendors; market-share takeaway.

Terms to know:

Kunpeng Huawei's family of ARM-based server processors.

ARM64 A processor architecture (as opposed to the more familiar x86 used by Intel/AMD) — power-efficient, and what Kunpeng chips use.

TaiShan / Inspur / XFusion Chinese server manufacturers that build rack servers certified to run Kunpeng processors and openEuler.

YMTC / ZhiTai Yangtze Memory Technologies Corp — China's domestic NAND flash memory manufacturer; ZhiTai is its consumer/enterprise SSD brand.

openEuler Huawei's open-source Linux distribution, built natively for ARM64.

GbE Gigabit Ethernet — wired networking speed, e.g. "10 GbE" = 10 gigabits per second.

Let Me Explain

So what does an all-Chinese hardware stack actually look like? On the processor side: Huawei's Kunpeng chips — ARM-based, with their newest generation shipping up to a hundred ninety-two cores per chip. Server chassis from Huawei's TaiShan line, or from Inspur and XFusion — all of them ship servers certified for Kunpeng and openEuler. Storage from YMTC, under their ZhiTai brand — that's China's homegrown 3D NAND flash memory manufacturer, with their newest drives hitting ten to fifteen gigabytes a second. And the operating system: openEuler, Huawei's open-source Linux, which already holds over half the market for newly deployed server operating systems in China. This isn't a fringe, unproven stack — Kunpeng-based servers already hold over twenty percent of China's server market. This is mainstream infrastructure we can simply choose to use.

Slide 22 — A concrete Bill of Materials [1:05 | running total 21:00]

On screen: Parts list table for a 100k-scale node.

Terms to know:

BOM Bill of Materials — the itemised parts list for building something.

PSU Power Supply Unit.

RDIMM Registered DIMM — an enterprise-grade memory module (more reliable, supports larger capacities than consumer memory).

PDU Power Distribution Unit — the rack-mounted power strip that feeds all the servers in a rack.

2U A unit of rack server height ("2U" = 2 rack units, about 3.5 inches).

Let Me Explain

Putting that into an actual parts list for a hundred-thousand-user-scale machine: a two-U rack server chassis from TaiShan or XFusion with dual power supplies, one or two Kunpeng processors, two hundred fifty-six gigabytes of ECC memory, a small mirrored boot drive, and the real workhorse — six NVMe drives in RAID ten for the database, giving us five to eight terabytes of usable, protected space. Dual ten-or-twenty-five-gigabit networking, openEuler as the operating system, and proper rack-level UPS support. Exact pricing moves with the currency exchange rate, so treat this as a specification to take to a reseller for a quote, not a fixed price.

6 Section V — Software & the Skills We Need

Slide 23 — Divider: Software & Skills [0:10 | running total 21:10]

Let Me Explain

Now — the software running on top of all that hardware, and the skills our team needs.

Slide 24 — The software stack: what changes [1:20 | running total 22:30]

On screen: "Stays the same" vs. "New, Chinese, and worth knowing".

Terms to know:

openGauss Huawei's own open-source fork of PostgreSQL, built for distributed, petabyte-scale workloads.

TiDB An open-source distributed SQL database from PingCAP, wire-compatible with MySQL, that scales horizontally across many machines.

OceanBase Ant Group's distributed database, built for financial-grade transaction processing at extreme concurrency.

OLTP Online Transaction Processing — the kind of workload made of many small, fast read/write transactions (as opposed to big analytical queries).

Let Me Explain

Here's the reassuring part: the database layer doesn't need to change. PostgreSQL is what Supabase runs today, and what OLK already depends on — open-source, portable, proven. Our whole application layer is already free to redeploy anywhere. But it's worth knowing what China's ecosystem offers if we ever outgrow a single machine: openGauss, which is actually Huawei's own fork of PostgreSQL, built for distributed, petabyte-scale workloads. TiDB, from a company called PingCAP, which speaks the same wire protocol as MySQL but scales horizontally. And OceanBase, from Ant Group, built for financial-grade transaction processing at extreme concurrency. Because openGauss is a PostgreSQL fork, that migration path exists — but we don't need it urgently. We'd only reach for something like this once a single well-specced machine, like our pilot, stops being enough — likely well past a hundred thousand users.

Slide 25 — Supabase self-hosting [1:05 | running total 23:35]

On screen: Three bullets on what Supabase actually is; the "headline point" takeaway.

Terms to know:

PostgREST Software that automatically turns a PostgreSQL database into a web API.

GoTrue Supabase's authentication service (sign-up, login, sessions).

Docker A tool for packaging software (and everything it needs to run) into portable containers that run identically on any machine.

Apache / MIT license Open-source software licenses that permit free use, modification, and redeployment — including for our own commercial or community use.

Let Me Explain

This is the headline point of the whole talk, honestly. Supabase itself is just a thin, open-source layer sitting on top of Postgres — a piece called PostgREST that turns the database into an API, GoTrue for authentication, a realtime engine, a storage API, and a management dashboard called Studio. All open-source, all licensed permissively, all packaged as Docker containers. We already run this exact stack locally for development. Running it in production, for real, is the same containers — just pointed at our own database, on our own machine, behind our own domain name. We do not need to rewrite Open Library Kashmir to make it sovereign. We need to redeploy it — first onto a VPS, then onto our own hardware.

Slide 26 — Skills matrix [1:05 | running total 24:40]

On screen: Table: skill, why it matters, whether we have it today.

Terms to know:

DBA Database Administrator — the role responsible for keeping a database backed up, tuned, and running reliably.

Firewalling Configuring network rules that control what traffic is allowed in and out of our servers.

Let Me Explain

None of this runs itself — here's what our team needs to build. Linux and openEuler administration, since everything runs on it. Docker and container orchestration — we're already good here. PostgreSQL database administration — backups, tuning, replication — we're partway there. Networking and firewalling for shutdown-resilient connections, security practices like patching and encryption, and basic facilities knowledge for power and cooling — those three, honestly, we don't have yet. This is as much a call for volunteers tonight as it is a call for hardware budget. Every gap in that "no" column is a role somebody in this room could take on.

Slide 27 — Backups & disaster recovery [1:05 | running total 25:45]

On screen: The 3-2-1 rule adapted for us; encryption takeaway.

Terms to know:

3-2-1 rule A standard backup principle: 3 copies of data, on 2 different media/locations, with 1 copy stored offsite.

OSS / COS Alibaba's "Object Storage Service" and Tencent's "Cloud Object Storage" — cheap, durable cloud storage for files, ideal for encrypted backup archives (not for hosting a live database).

Let Me Explain

Backups follow a simple, well-tested rule: three-two-one. Three copies of the data — the live production copy plus two backups. Two different kinds of media or locations — say, a local snapshot plus a separate physical disk or machine. And one copy offsite — either a different city in Jammu and Kashmir, or an encrypted bucket on a Chinese cloud storage service like Alibaba OSS or Tencent COS, used strictly as cold backup, never as the live host. Everything is encrypted before it ever leaves our building. Used that way, an offsite Chinese cloud bucket carries none of the sovereignty compromise of actually hosting live traffic there.

7 The Hope

Slide 28 — Divider: The Hope [0:15 | running total 26:00]

Let Me Explain

And that brings me to the part I actually want to end on — the hope.

Slide 29 — The proposal: one machine, real capacity [1:35 | running total 27:35]

On screen: Pilot spec bullets, "why this scope", and the headroom calculation row.

Let Me Explain

Here's the ask, in concrete terms. A pilot serving ten to twenty thousand users, off a single, well-specced machine: five terabytes of usable NVMe storage in RAID ten, a hundred twenty-eight gigabytes of ECC memory, a Kunpeng-class processor, openEuler, and our own PostgreSQL and self-hosted Supabase stack — Chinese hardware and hosting from the virtual machine all the way down to the silicon. This is deliberately modest: small enough to fund and run with the volunteer skills we can realistically build this year, but large enough to be a genuinely real platform, not a demo. It proves every single claim I've made tonight before we ever ask for datacenter-scale money. And the math backs it up — twenty thousand users at roughly five megabytes each is only about a hundred gigabytes of active data. Against five terabytes, that's fifty times headroom. Years of growth, every book cover, every backup, without touching the hardware again.

Slide 30 — The road from here (closing) [1:35 | running total 29:10]

On screen: Three-phase roadmap diagram; the closing ask; "Our data. Our servers. Our region. Let's begin."

Terms to know:

N+1 redundancy Having one more of a critical component than the minimum needed, so a single failure (of a power supply, a link, a cooling unit) doesn't take the system down.

Let Me Explain

So here's the road ahead, in three phases. Phase one, this year and next: this exact pilot, on Chinese VPS or colocation, serving ten to twenty thousand users — proving the model works. Phase two, the year after: our own room, with real redundant power and cooling. Phase three, from twenty-twenty-eight onward: a multi-site data center here in Kashmir, serving well past a hundred thousand users, as a genuine regional model. What I'm asking for tonight is simple: a funding commitment for the phase-one parts list, volunteers against the skills we just went through, and eventually, a room we can call our own. This is buildable in months, not decades. It starts the moment we agree to start. Our data. Our servers. Our region. Let's begin. Thank you.

Appendix: Full glossary, A–Z

3-2-1 rule Backup principle: 3 copies, on 2 media/locations, 1 offsite.

ARM64 A power-efficient processor architecture used by Kunpeng and most mobile and modern server chips, as an alternative to x86.

Auto-transfer switch Hardware that automatically switches power source from grid to generator without human intervention.

BOM Bill of Materials — an itemised hardware parts list.

Capex Capital expenditure — upfront spending on physical assets.

CDN Content Delivery Network — cached copies of content served from locations closer to the user.

CLOUD Act US law compelling US companies to hand over data on request regardless of where the server is physically located.

Colocation ("colo") Renting rack space/power/network in someone else's facility while owning the server hardware yourself.

Concurrent users Users actively using a system at the same moment.

CVM Cloud Virtual Machine — Tencent Cloud's rentable virtual server product.

DBA Database Administrator.

DIMM Dual In-line Memory Module — a physical memory stick.

Docker A tool for packaging software into portable, self-contained containers.

ECC RAM Error-Correcting Code memory — detects and fixes single-bit errors.

ECS Elastic Compute Service — Alibaba's/Huawei's rentable virtual server product.

FM-200 / Novec Clean-agent fire-suppression gases safe for use around live electronics.

Free-air / economizer cooling Cooling a server room using outside air directly instead of mechanical air-conditioning.

Gbps / Mbps Gigabits/megabits per second — network speed units.

GbE Gigabit Ethernet — wired network speed (e.g. 10 GbE = 10 Gbps).

GoTrue Supabase's authentication microservice.

Hyperscaler A cloud provider operating at massive global scale.

ICP license Mainland Chinese filing required only for websites physically hosted inside mainland China; avoided by using a Hong Kong/Singapore region.

IOPS Input/Output Operations Per Second — how many read/write operations a storage device can handle per second.

KPDCL Kashmir Power Distribution Corporation Limited.

Kunpeng Huawei's family of ARM-based server processors.

Latency Round-trip delay for data to travel between two points, in milliseconds.

Lighthouse Tencent Cloud's simplified all-in-one VPS product.

Mirroring Writing identical data to two drives simultaneously (RAID 1's basis).

N+1 redundancy Having one spare unit beyond the minimum needed so a single failure doesn't cause an outage.

Next.js / React The open-source web framework and library OLK's front end is built with.

NVMe A storage protocol letting SSDs communicate over high-speed PCIe lanes.

OceanBase Ant Group's distributed, financial-grade database.

OLTP Online Transaction Processing — many small, fast transactions.

openEuler Huawei's open-source, ARM64-native Linux distribution.

openGauss Huawei's open-source PostgreSQL fork for distributed workloads.

OSS / COS Alibaba Object Storage Service / Tencent Cloud Object Storage — cloud file storage, suitable for encrypted backups only.

Parity Derived data stored alongside real data, allowing reconstruction of a failed drive.

Pay-as-you-go Billing by actual usage rather than a fixed upfront commitment.

PDU Power Distribution Unit — rack-mounted power strip.

PostgREST Software that exposes a PostgreSQL database as a web API.

PostgreSQL The open-source relational database engine underlying OLK and Supabase.

PSU Power Supply Unit.

RAID Redundant Array of Independent Disks — combining drives for redundancy and/or speed.

RDIMM Registered DIMM — enterprise-grade server memory.

RDSS Revamped Distribution Sector Scheme — India's power-sector reform program.

SATA SSD A solid-state drive using the older, slower SATA interface (capped ∼550 MB/s).

`shared_buffers` A PostgreSQL setting for how much RAM it reserves as its own cache.

Striping Splitting data across multiple drives for parallel, faster read/write.

Supabase Open-source platform bundling Postgres with auth, storage, and an API layer.

TaiShan / Inspur / XFusion Chinese server manufacturers building Kunpeng/openEuler-certified rack servers.

TiDB PingCAP's open-source, MySQL-wire-compatible distributed SQL database.

UPS (online double-conversion) A battery backup providing zero-delay switchover when grid power fails.

VPS Virtual Private Server — a rented slice of a shared physical machine.

WAL Write-Ahead Log — PostgreSQL's crash-safety mechanism.

YMTC / ZhiTai China's domestic NAND flash manufacturer and its SSD brand.

2U A rack server height unit (2 rack units $\approx$ 3.5 inches).